The decision trail behind the retained dual-3090 profile

Fit, client behavior, long-prefill decay, speculation, correctness and restoration each changed the next experiment.

01 / REAL WORKLOAD

Start from the client

Resume an existing task
Images, tools and follow-ups
Several sessions, one service

02 / FIT AND STARTUP

Check the GGUF path

llama.cpp Q4_K_M trials
Near-1M configured limits
Allocation + tiny-input logs

03 / LONG HISTORY

Measure long-prefill decay

vLLM 0.23 W4A8 PP2
520k prefill slowed late
High hits, slow next-turn decode

04 / SERVING CHOICES

Probe layout and speculation

SGLang TP/PP; dynamic gate
MTP, DFlash2 and DFlash8
Precision and CUDA graph modes

05 / LOCAL CONTROLS

Check matched candidates

Native 0.28 and 0.30 trials
Legacy verification comparisons
Custom branch later retired

06 / RETAINED PROFILE

Select public upstream code

Native vLLM + DFlash2
Nine unchanged public patches
18.6k warm median: 153 tok/s

07 / RECOVERY EVIDENCE

Restore and reconstruct

Archive restores: 151 / 146
Separate assembly: 145 tok/s
One text-prefix disk reload

08 / REUSABLE DEPLOYMENT

Share a recoverable recipe

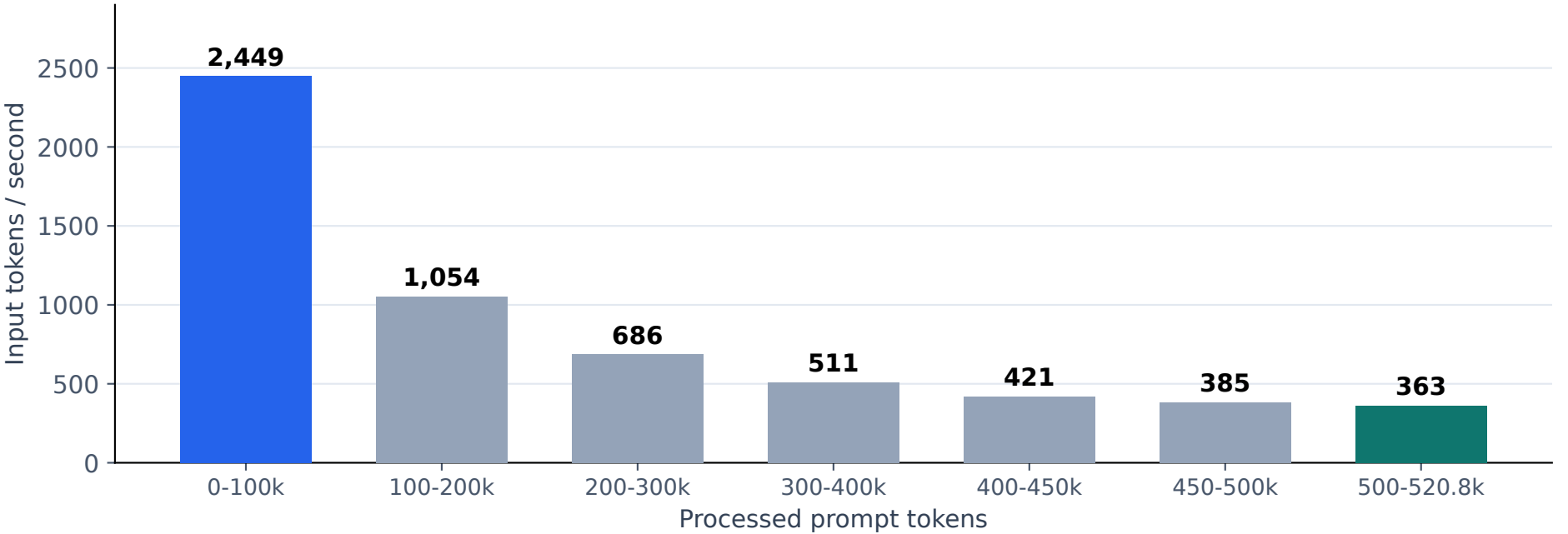
Several clients, one server
Disk-prefix reuse after reload
Pinned setup + timing records

Campaigns overlap. Each result is linked to its configuration and original measurement record.

Measured headline: 18,649 actual input tokens with a warm prefix; output rates include reasoning.

Prefill across a real 520k-token history

vLLM 0.23 W4A8 PP2 | Median completed-batch prefill rate by processed-token range

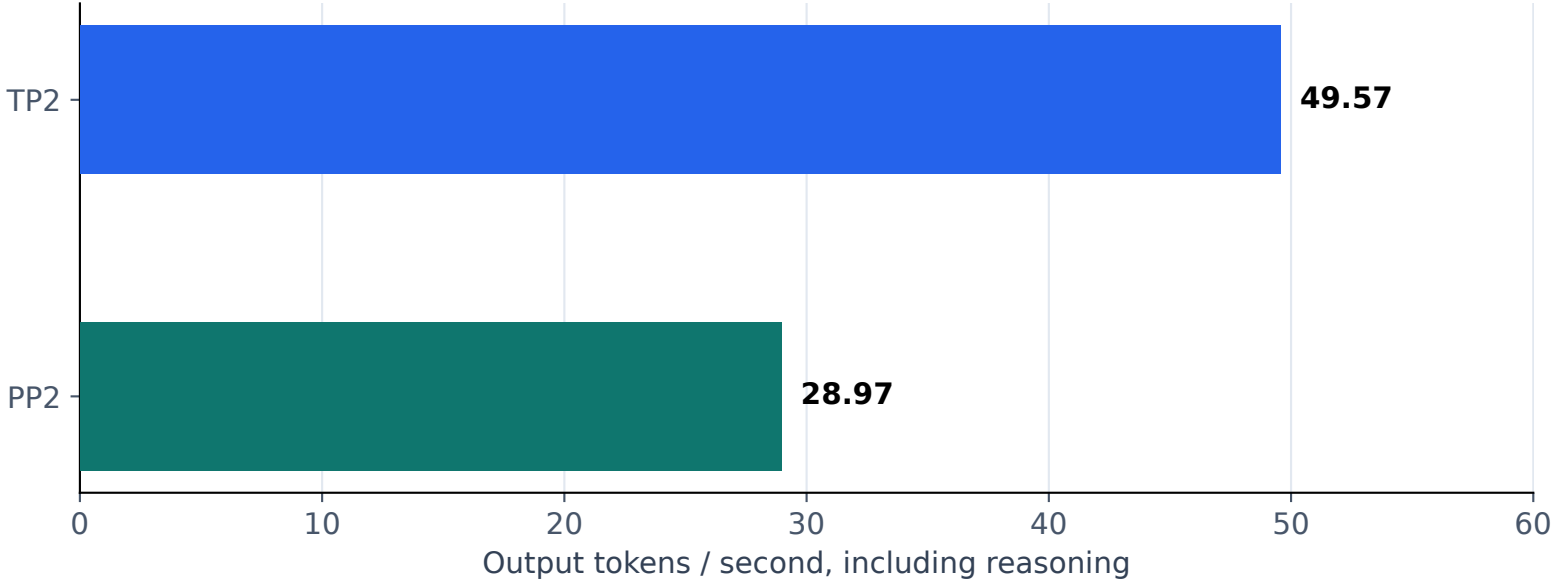


Actual input: 520,761 tokens. First batch excluded. Bins contain 31, 32, 32, 32, 16, 16 and 8 batches.

The separate 3,437 tok/s early sampling window uses a different metric. Source: S038 in benchmarks/history.

Same first paper prompt, different SGLang layouts

SGLang 0.5.19 + MTP | 5,224 input / 4,096 output tokens | One run per layout

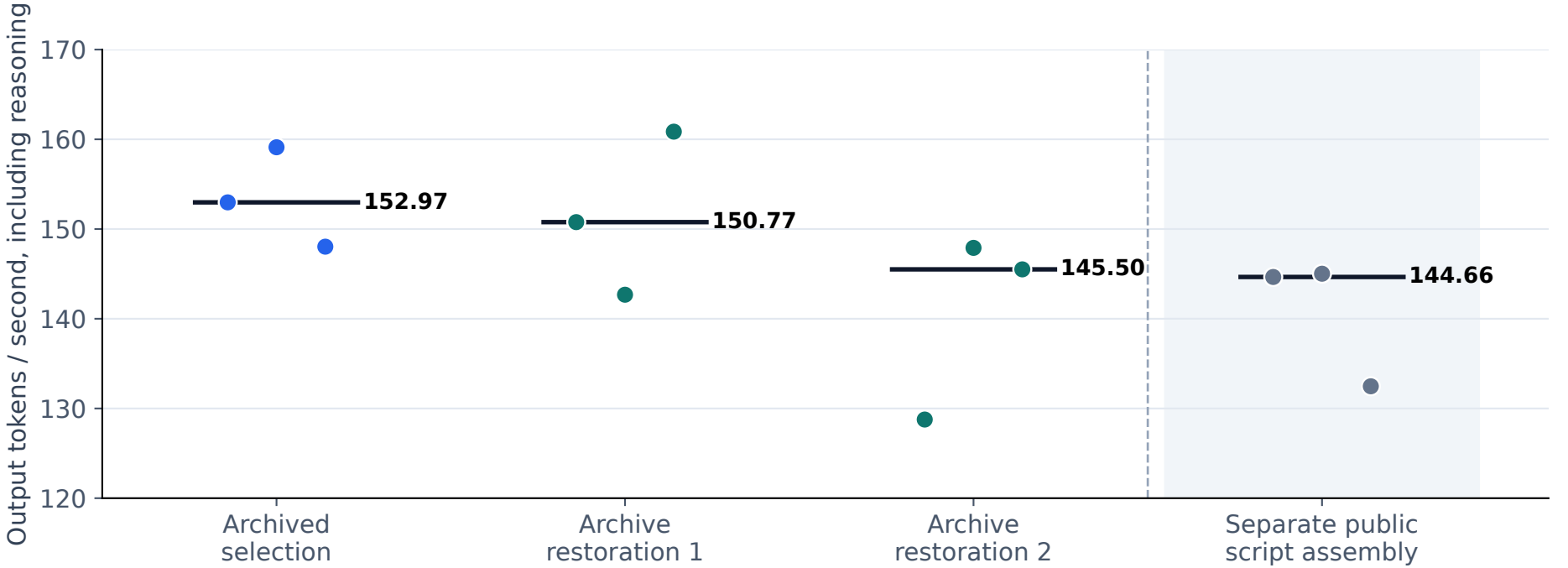


PP2: AWQ W4A16, FP8 KV, historical backports. Full TP2 launch recovery is incomplete.

First API phase only; later contexts diverge. Other configuration effects are not isolated. Source: S016.

Repeated measurements of the retained profile

18,649 input / 1,024 output tokens per stream | Three observations in each group



Dots: individual capped runs, including reasoning. Black segments: group medians. Three observations per group.

First three groups: 17,920 prefix hits. Separate assembly: 16,128 hits, fixed 10.5 GiB KV/card, smaller shared pool.

Sources: benchmarks/results.json, history D008/D009/D010, and assembly-validation-20261017.json.